

Tea Leaf Disease Classification:

VGG16 vs. ResNet50 - Performance Comparison Report

1. Introduction

This report offers a comparative evaluation of two convolutional neural network (CNN) architectures—VGG16 and ResNet50 implemented for a three-class classification task focused on tea leaf diseases. The target classes include Algal Leaf, Brown Blight, and White Spot. Both models utilized transfer learning with pre-trained weights from ImageNet and incorporated data augmentation techniques to enhance a limited dataset of leaf images.

2. Training History

Figure 1 illustrates the cross-entropy loss and classification accuracy across 25 training epochs for both models. ResNet50 demonstrated faster convergence and consistently lower validation loss, stabilizing around epoch 12, while VGG16 exhibited higher variance and a persistently elevated validation loss throughout training.

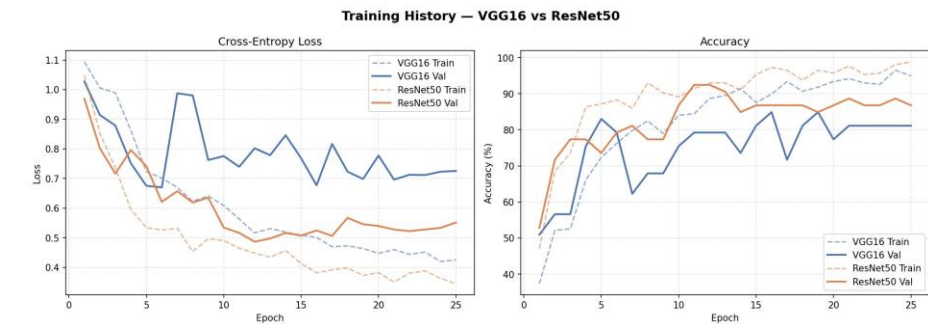


Figure 1. Training and validation loss/accuracy curves - VGG16 vs. ResNet50 (25 epochs).

3. Test Set Performance

Table 1 summarizes macro-averaged performance metrics on the held-out test set. ResNet50 outperforms VGG16 across all four metrics, achieving a +5.17-percentage point gain in test accuracy. Values in bold indicate the superior result per metric.

Table 1. Overall Test Performance (macro-averaged).

Metric	VGG16	ResNet50	Δ (ResNet50 – VGG16)
Test Accuracy	82.76%	87.93%	+5.17 pp
Precision (macro)	83.7%	88.3%	+4.6 pp
Recall (macro)	82.8%	87.7%	+4.9 pp
F1-Score (macro)	83.0%	87.9%	+4.9 pp

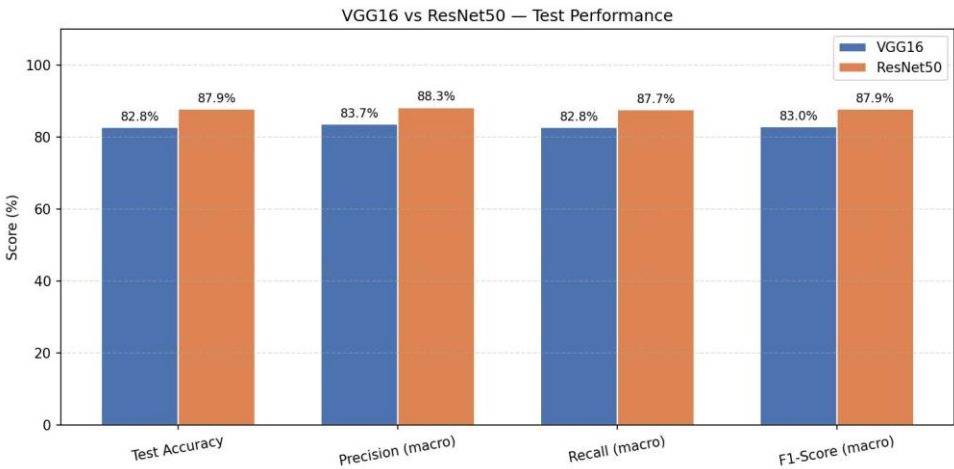


Figure 2. Side-by-side comparison of test accuracy, precision, recall, and F1-score.

4. Per-Class Analysis (Confusion Matrices)

Table 2 and Figure 3 provide insights into the per-class recall metrics derived from the confusion matrices of the test set. ResNet50 exhibits a significant enhancement in recall for Brown Blight (+16.67 percentage points) and White Spot (+9.09 percentage points). Meanwhile, VGG16 demonstrates superior recall for Algal Leaf (94.44% compared to 83.33%), indicating its ability to capture algal texture patterns more effectively. However, it struggles with Brown Blight, the most challenging class, misclassifying 27.78% of samples as White Spot.

Table 2. Per-Class Recall from Confusion Matrices.

Class	VGG16 Recall	ResNet50 Recall	Δ
Algal Leaf	94.44%	83.33%	−11.11 pp
Brown Blight	72.22%	88.89%	+16.67 pp
White Spot	81.82%	90.91%	+9.09 pp

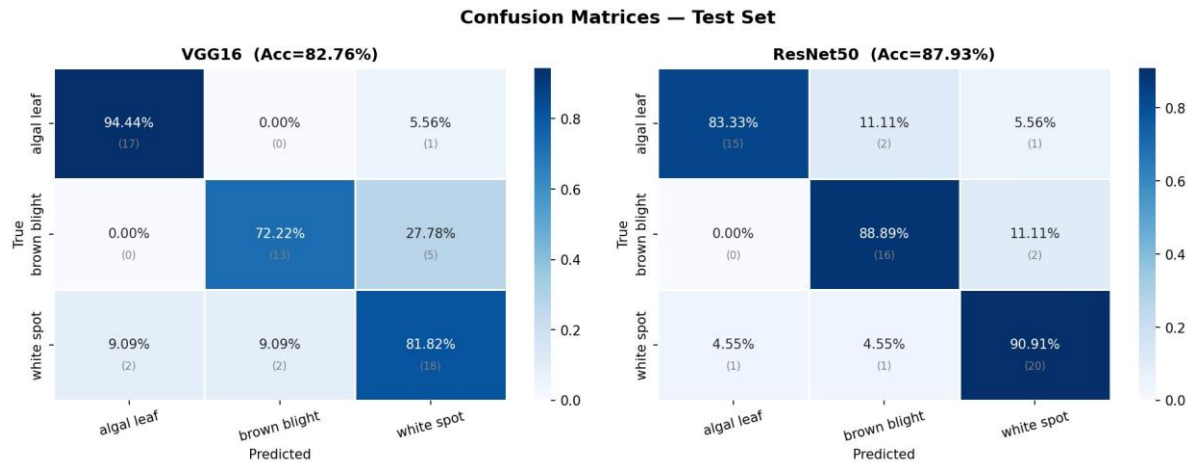


Figure 3. Confusion matrices for VGG16 (left, Acc=82.76%) and ResNet50 (right, Acc=87.93%) on the test set.

5. Discussion

ResNet50 consistently outperforms VGG16 across aggregate metrics, primarily due to its residual learning mechanism, which enhances the extraction of more discriminative features from a limited dataset. A notable weakness of VGG16 is its performance in classifying Brown Blight, where the visual similarities between partially browned leaves and other disease patterns often lead to confusion. In contrast, ResNet50’s residual connections seem to effectively capture more subtle textural gradients, allowing for a more reliable distinction of Brown Blight.

Interestingly, VGG16 achieves a higher recall for Algal Leaf classification (94.44%), likely due to the simpler visual signature of greenish discoloration, which the sequential convolutional structure can more easily identify. Furthermore, the persistent gap between training and validation accuracy in both models, especially in ResNet50, where training accuracy reaches approximately 99%, highlights the issue of overfitting, likely caused by the small dataset size. Addressing this may require additional regularization or further data collection. is recommended.

6. Conclusion

ResNet50 emerges as the most effective architecture for the classification of tea leaf diseases, achieving a test accuracy of 87.93% compared to VGG16’s 82.76%. This significant improvement is reflected across all macro-averaged metrics. A key advantage of ResNet50 is its ability to accurately classify Brown Blight, a category where VGG16 struggles. Future research should focus on expanding data collection, implementing class-balanced sampling, and utilizing Grad-CAM visualization to enhance model interpretability and robustness.